VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI

A PROJECT PHASE-1 REPORT ON

"Skill2Job: AI-Driven Placement Coordination and Skill Mapping System"

Submitted in partial fulfillment for the award of Degree of,

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

By

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ATME COLLEGE OF ENGINEERING

(Affiliated to Visvesvaraya Technological University, Belagavi &
Approved by AICTE, New Delhi. Recognized by Government of Karnataka.)
Accredited by NBA and NAAC with A+ Grade
13th Kilometer Mysore-Kanakapura-Bangalore Road, MYSORE-570028 Karnataka

2025 – 2026

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CERTIFICATE

This is to certify that the Project Phase 1 entitled **"Skill2Job: AI-Driven Placement Coordination and Skill Mapping System"** has been successfully completed by

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the bonafide students of **Department of Computer Science and Engineering , ATME College of Engineering,** in partial fulfillment for the award of BACHELOR OF ENGINEERING in **DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING** of the **VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI,** during the year 2025–2026. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project phase 1 report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the Bachelor of Engineering Degree.

Dr. Shilpa B L
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Declaration

We,

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hereby declare that the Project phase 1 report entitled, **Skill2Job: AI-Driven Placement Coordination and Skill Mapping System** is completed and written by us under the supervision of our guide **Dr Shilpa B L, Associate Professor , Department of Computer Science and Engineering, ATME College of Engineering, Mysore-570028,** in partial fulfillment of the requirements for the award of the degree **BACHELOR OF ENGINEERING** in **DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING** of the **VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI,** during the academic year 2025-2026.The Project phase 1 report is original and it has not been submitted for any other degree in any university.

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ABSTRACT

The rapid growth of the job market and increasing competition among students has made placement coordination a complex and time-consuming task for educational institutions. Traditional placement systems rely heavily on academic scores such as CGPA and basic eligibility criteria, which often fail to reflect the actual skill level of students. As a result, many capable students remain unmatched with suitable job roles, and placement officers spend excessive time manually filtering candidates.

The proposed project, Skill2Job, is an Artificial Intelligence (AI)-driven platform designed to automate placement coordination and skill mapping processes. The system analyzes student profiles, identifies technical skills, generates professional resumes automatically, and recommends suitable job roles based on skill compatibility. Additionally, the system identifies skill gaps and suggests relevant courses or learning paths to improve employability.

The system benefits students, placement officers, and institutions by providing faster candidate shortlisting, improved accuracy in job-role matching, and better placement outcomes. The integration of machine learning algorithms ensures efficient skill-based filtering and recommendation processes. Overall, the Skill2Job platform enhances placement efficiency, supports career growth, and bridges the gap between academic learning and industry requirements.

Keywords: *Artificial Intelligence, Placement Coordination, Skill Mapping, Natural Language Processing, Machine Learning, Resume Generation, Job Recommendation.*

LIST OF TABLES

LIST OF FIGURES

TABLE OF CONTENTS

CHAPTER 1

INTRODUCTION

1.1 Introduction

In today's rapidly evolving technological environment, the job market has become highly competitive. Employers are increasingly focusing on practical skills, project experience, and domain knowledge rather than solely relying on academic performance. Students are expected to demonstrate proficiency in various technologies and tools to secure suitable employment opportunities.

Most educational institutions still depend on traditional placement methods that involve manual data entry, resume verification, and eligibility filtering based on academic performance. These methods are inefficient and often lead to mismatches between student capabilities and company requirements. Placement officers face challenges in managing large volumes of student data, resulting in delays and inaccuracies.

Artificial Intelligence has emerged as a powerful solution for automating complex decision-making processes. AI-based systems can analyze large datasets, identify patterns, and provide accurate recommendations within a short period. Integrating AI into placement systems enables intelligent skill analysis, resume generation, and job-role matching.

The Skill2Job system aims to modernize campus placement processes by incorporating AI-driven techniques. The platform provides automated resume generation, skill analysis, job-role recommendation, and skill-gap identification. By combining intelligent algorithms and structured data processing, the system ensures efficient placement coordination and improved employability outcomes.

1.2 Problem Statement

Placement coordination in many colleges is still handled manually, which makes it difficult to match student skills with company requirements. The process mainly focuses on basic criteria like CGPA, without properly evaluating or tracking student technical skills. As a result, students often remain unaware of the skills needed for specific job roles.

Manual filtering of eligible candidates also consumes a great deal of time for placement officers. This approach does not consider technical skills, project experience, or domain expertise, which are critical for job selection. The lack of an intelligent system results in delayed shortlisting, inaccurate job-role matching, and limited placement opportunities. Hence, there is a need for an intelligent system that can automatically map skills and provide suitable job recommendations.

1.3 Motivation and Objectives of the Project

The motivation behind Skill2Job arises from the evident gap between student skill sets and industry expectations. The need to modernize campus placement with AI-powered tools has driven this project.

The main objectives of Skill2Job are:

- To develop an automated resume generation system based on student profile and key skills.
- To develop an AI-based placement coordination system and map student skills with company job requirements.
- To identify skill gaps for students and provide course recommendations based on skill profiles.
- To develop a system that enables placement officers to shortlist eligible candidates and analyze placement success rates with improved accuracy and reduced processing time.

1.4 Existing System & Drawbacks

The existing placement systems used in many institutions are primarily manual. Student data is collected through forms and stored in spreadsheets. Placement officers verify eligibility criteria manually and send selected resumes to companies.

Limitations of the Existing System:

- Time-consuming data processing and manual resume verification.
- Limited skill evaluation — only CGPA and basic eligibility are checked.
- Increased workload for placement officers managing large datasets.
- High possibility of human errors in candidate shortlisting.
- No mechanism for skill-gap identification or learning path suggestions.

1.5 Proposed Solution & Advantages

The proposed Skill2Job system is an AI-driven platform designed to automate placement coordination and skill mapping processes. The system collects student data, analyzes skill sets, generates professional resumes, and recommends suitable job roles.

Key advantages of the proposed system:

- Automated resume generation based on student profile and skills.
- Intelligent skill mapping with accurate job recommendations using ML algorithms.
- Identification of skill gaps and suggestion of relevant courses or learning paths.
- Reduced manual workload for placement officers with faster candidate shortlisting.
- Real-time insights into student performance and placement analytics.

CHAPTER 2

LITERATURE SURVEY

Several research studies have explored the application of Artificial Intelligence in career guidance and placement systems. The following literature survey highlights important contributions and their limitations.

[1] Selvaraj et al. (2025) — AI-Driven Intelligent Career Mapping and Skill-based Job Role Predicting Career Guidance System This paper proposes a career guidance system that uses LightGBM to predict suitable job roles based on student skills, with SMOTE applied to handle class imbalance in the dataset. It achieved a strong accuracy of 90.18%, making it effective for skill-based career prediction. However, the study is limited by a small dataset (~500 records) and relies on self-reported skills without deep learning models.

[2] El-Khalili et al. (2025) — AI-Powered Career Guidance System This work combines a Random Forest classifier with the Big Five personality test to recommend career paths, achieving 87.3% accuracy and an AUC-ROC of 0.99. It also integrates VR-based career exploration to give students an immersive experience of potential job environments. The main limitation is its reliance on a Western-context dataset, reducing applicability to diverse regional settings.

[3] Mirajkar et al. (2024) — Skill Recommendation System and Resume Analysis using AI The authors developed an NLP-based resume parsing system using Streamlit that extracts skills, scores resumes, and visualizes career field suitability. It focuses on resume-level analysis and provides skill recommendations based on predefined category mappings. The system lacks job-role matching capabilities and does not report quantitative accuracy metrics.

[4] Gangoda et al. (2024) — Resume Ranker: AI-Based Skill Analysis and Skill Matching System This system uses SpaCy, KeyBERT, and Word2Vec with cosine similarity to rank candidates by matching resume skills against job requirements, achieving F1 scores of 0.89–0.95. It provides a data-driven shortlisting mechanism useful for recruiters handling large applicant pools. Limitations include dependency on LinkedIn data and specificity to the Japan labor market, with limited skill-gap analysis.

[5] Balakumar et al. (2024) — Impact of AI-Driven Automation on Job Displacement and Skill Development This study applies Gradient Boosting combined with Explainable AI (SHAP) to predict job displacement risk due to automation, achieving 98% accuracy and 97% precision. The use of SHAP values adds transparency by explaining which factors contribute most to displacement predictions. However, it is purely predictive and does not offer actionable skill development recommendations or placement coordination features.

[6] Jawhar et al. (2024) — AI-Powered Customized University and Career Guidance The paper presents an NLP-based chatbot that profiles students based on personality and academic data to suggest suitable universities and career paths, with 85% student satisfaction. It personalizes guidance by integrating multiple data inputs including extracurricular interests and academic performance. The system is limited to university selection and does not address skill-gap analysis, resume generation, or job-matching components.

[7] Baklarova et al. (2025) — AI-Based Advisory System for Career Guidance This research develops a knowledge-based advisory system that uses AI decision trees to provide career field suggestions, reaching 82% accuracy in career field recommendations. The rule-based architecture makes the system interpretable but limits its adaptability to dynamic, real-world skill markets. It lacks resume generation and job-matching modules, making it unsuitable as a complete placement coordination solution.

[8] Rakshitha et al. (2025) — An AI-Driven Framework for Personalized Career Path Recommendations The authors propose a hybrid recommendation engine combining collaborative filtering and content-based approaches to deliver personalized career roadmaps, achieving 88% user satisfaction. The system tailors suggestions based on individual student profiles and historical career data from similar users. Key limitations include no automated resume generation and the absence of a placement officer dashboard or institutional integration.

[9] Prathyusha et al. (2025) — Skill Mentor: AI-Driven Student Skill Enhancement and Placement Prediction Framework This framework uses Random Forest and SVM to predict placement outcomes and suggest skill enhancement paths, achieving 91% placement prediction accuracy. It bridges skill development and placement readiness by identifying gaps between student competencies and industry expectations. The system is limited to a single institution's dataset and does not include an admin dashboard or automated resume generation.

[10] Kumar & Singh (2023) — Machine Learning Techniques for Resume Classification and Candidate Recommendation The paper applies SVM combined with TF-IDF feature extraction to classify resumes and recommend candidates, achieving 93% accuracy in candidate ranking. It demonstrates the effectiveness of traditional ML techniques for automating the early-stage recruitment screening process. The system uses a static skill taxonomy with no dynamic skill-gap identification or course recommendation feature.

Table 2.1: Literature Survey Summary

Author & Year	Paper Title	Method / Result	Limitations / Gap
Selvaraj et al. (2025)	AI-Driven Intelligent Career Mapping and Skill-based Job Role Predicting Career Guidance System	LightGBM career prediction using student skills and SMOTE balancing; accuracy 90.18%	Small dataset (~500 records), self-reported skills, no deep learning, limited tracking
El-Khalili et al. (2025)	AI-Powered Career Guidance System	Random Forest with Big Five personality test; VR career exploration; 87.3% accuracy, AUC-ROC 0.99	Western-context dataset, limited regional data, multi-class Gradient Boosting issues
Mirajkar et al. (2024)	Skill Recommendation System and Resume Analysis using AI	NLP resume parsing via Streamlit; skill extraction, resume scoring, career field visualization	Resume-focused only, predefined skill categories, no job matching, no accuracy metrics reported
Gangoda et al. (2024)	Resume Ranker: AI-Based Skill Analysis	SpaCy, KeyBERT, Word2Vec with	LinkedIn dependency, Japan labor market-

Author & Year	Paper Title	Method / Result	Limitations / Gap
	and Skill Matching System	cosine similarity; F1 score 0.89–0.95	specific, limited skill-gap analysis
Balakumar et al. (2024)	Impact of AI-Driven Automation on Job Displacement and Skill Development	Gradient Boosting + Explainable AI (SHAP); 98% accuracy, 97% precision	Prediction-only focus, single Kaggle dataset, no actionable skill recommendations
Jawhar et al. (2024)	AI-Powered Customized University and Career Guidance	NLP-based chatbot with personality profiling; 85% student satisfaction in university and career recommendations	Limited to university selection only; no skill-gap analysis, resume generation, or job-matching component
Baklarova et al. (2025)	AI-Based Advisory System for Career Guidance	Knowledge-based advisory system using AI decision trees; 82% accuracy in career field suggestion	Rule-based approach limits adaptability; no resume generation or job-matching module integrated
Rakshitha et al. (2025)	An AI-Driven Framework for Personalized Career Path Recommendations	Collaborative filtering + content-based hybrid model; personalized roadmaps with 88% user satisfaction	No automated resume generation; lacks placement officer dashboard and institutional integration
Prathyusha et al. (2025)	Skill Mentor: AI-Driven Student Skill Enhancement and	Random Forest + SVM placement prediction; skill	No automated resume generation; limited to single-institution

Author & Year	Paper Title	Method / Result	Limitations / Gap
	Placement Prediction Framework	enhancement suggestions; 91% placement prediction accuracy	dataset; no admin dashboard
Kumar & Singh (2023)	Machine Learning Techniques for Resume Classification and Candidate Recommendation	SVM + TF-IDF for resume classification; 93% accuracy in candidate ranking and recommendation	Static skill taxonomy; no dynamic skill-gap identification or course recommendation feature

The above Table 2.1 highlight the importance of AI in career and placement systems. However, most existing systems focus on individual components such as resume analysis or career prediction. The proposed Skill2Job system integrates resume generation, skill mapping, job recommendation, and skill-gap identification into a unified platform, addressing the limitations identified in previous research.

CHAPTER 3

REQUIREMENT SPECIFICATION

3.1 Functional Requirements

- Student registration, login, and profile management.
- Automated professional resume generation based on student data.
- Skill analysis and mapping using NLP and machine learning.
- Job-role recommendation based on skill compatibility scores.
- Skill-gap identification and course recommendation for improvement.
- Admin module for managing student records, company data, and placement progress.
- Placement officer dashboard for shortlisting candidates and viewing analytics.

3.2 Non-Functional Requirements

- Performance: The system should process skill analysis and job matching within an acceptable response time even with large datasets.
- Scalability: The platform should handle multiple simultaneous users — students, admins, and placement officers — without degradation.
- Security: User data must be stored securely with authentication and authorization controls.
- Usability: The interface should be intuitive and user-friendly for students with varying levels of technical proficiency.
- Reliability: The system must maintain consistent accuracy in job-role matching and skill-gap analysis.
- Maintainability: The codebase should be modular and well-documented for future enhancements.

3.3 Technical Requirements (Hardware / Software)

Hardware Requirements:

- Processor: Intel Core i5 or higher (or equivalent)
- RAM: Minimum 8 GB

- Storage: Minimum 256 GB SSD
- Network: Stable internet connection for deployment and API access

Software Requirements:

Table 3.1 Technology Stack

Category	Technologies
Frontend	HTML, CSS, JavaScript, React (Optional)
Backend	Python, Flask / Node.js
Database	MySQL
AI/ML Techniques	Natural Language Processing (NLP), Machine Learning Algorithms

This table 3.1 illustrates the overall technology stack used in the system. It shows the interaction between frontend, backend, database, and AI modules. It provides a clear view of system implementation layers.

CHAPTER 4

PROPOSED METHODOLOGY

The methodology describes the sequence of steps followed in the Skill2Job system. The system is designed to work as an end-to-end automated placement coordination platform.

1. Student Registration — Students create an account and provide personal and contact details.
2. Profile Creation — Students enter academic information, technical skills, and project details.
3. Resume Generation — The system automatically generates a professional resume based on the entered data using predefined templates.
4. Skill Analysis — Machine learning algorithms (NLP-based) analyze and categorize the student's technical skills.
5. Job Matching — The system compares student skill vectors with job requirement vectors using cosine similarity or similar ML techniques.
6. Skill Gap Identification — Missing skills are identified by comparing the student's profile against job role requirements.
7. Recommendation Generation — Suitable job roles and relevant online courses are recommended to bridge identified skill gaps.

This methodology ensures that each student receives a personalized experience, and placement officers receive a ranked shortlist of candidates for each company requirement — all with minimal manual intervention.

CHAPTER 5

SYSTEM ARCHITECTURE

The system architecture consists of multiple interacting components that collectively perform placement coordination tasks. The major components are:

- Student Interface — Web-based front-end for students to register, manage profiles, generate resumes, and view job recommendations.
- Admin Interface — Dashboard for placement officers and administrators to manage student data, add companies, and monitor placement activities.
- Skill Analyzer Module — NLP-based engine that extracts and categorizes skills from student profiles.
- Resume Generator Module — Template-driven module that automatically creates professional resumes in PDF/DOCX format.
- Job Matching Engine — ML algorithm that computes compatibility scores between student skills and job requirements.
- Database Server — MySQL relational database storing student profiles, company data, job roles, and skill mappings.

The flow diagram below illustrates the end-to-end Student Job Matching Cycle of the Skill2Job system:

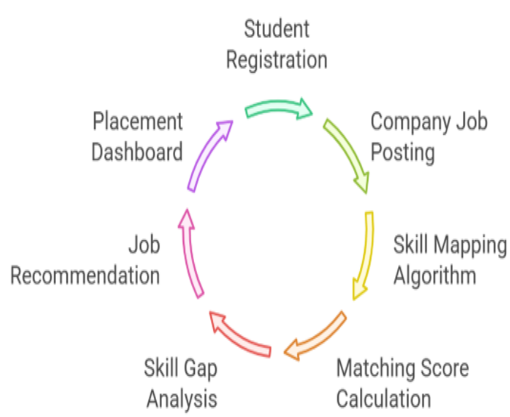

Figure 5.1: Student Job Matching Cycle – Flow Diagram

Figure 5.1 represents the architecture ensures secure data storage, efficient multi-module processing, and accurate real-time recommendation results. The modular design also supports future integration with corporate recruitment systems and online learning platforms.

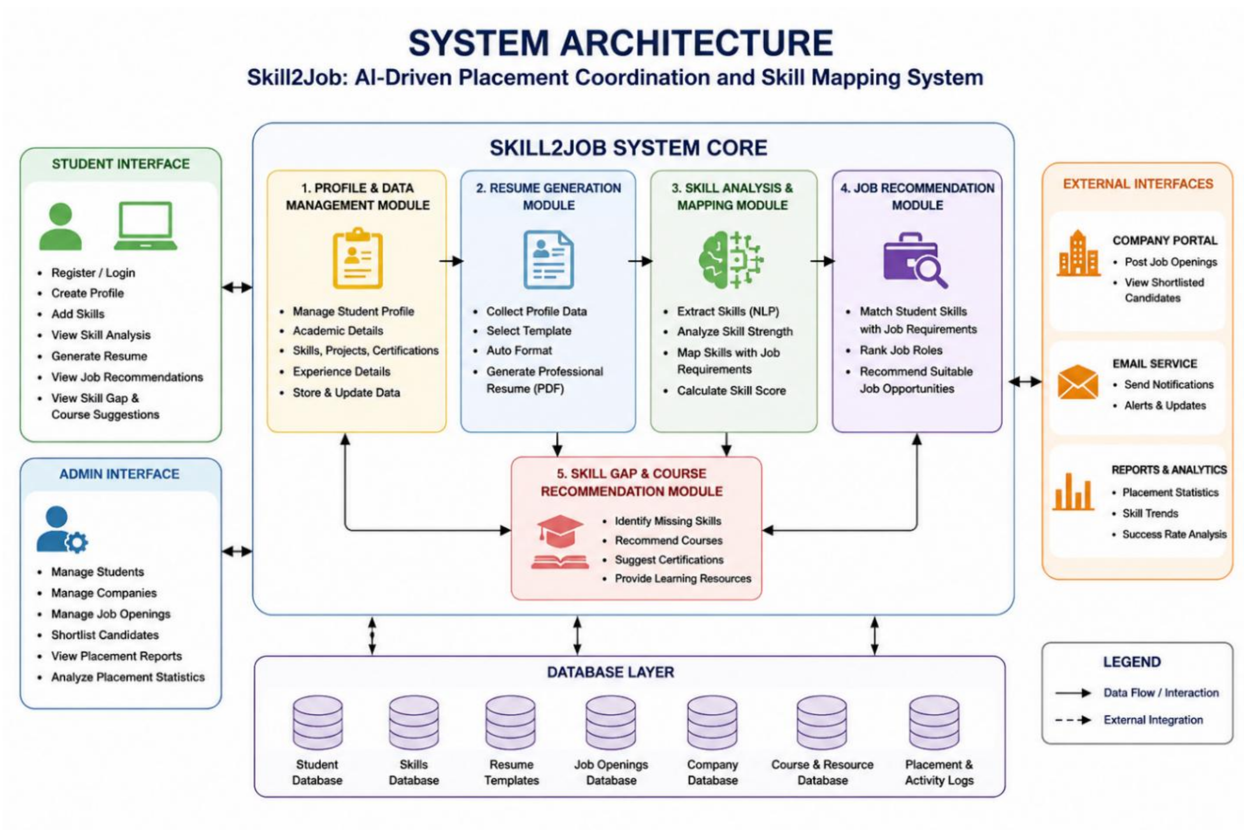

Figure 5.2: System Architecture

The above Figure 5.2 Skill2Job system architecture consists of five core processing modules — Profile & Data Management, Resume Generation, Skill Analysis & Mapping, Job Recommendation, and Skill Gap & Course Recommendation — all integrated within the central Skill2Job System Core. The Student Interface allows students to register, create profiles, generate resumes, and view personalized job recommendations and skill-gap suggestions, while the Admin Interface enables management of students, companies, job openings, and placement analytics.

Data Flow Diagram

The Data Flow Diagram (DFD) represents the overall functioning of the Skill2Job: AI-Driven Placement Coordination and Skill Mapping System. It illustrates how data flows between users, system modules, and databases across different levels.

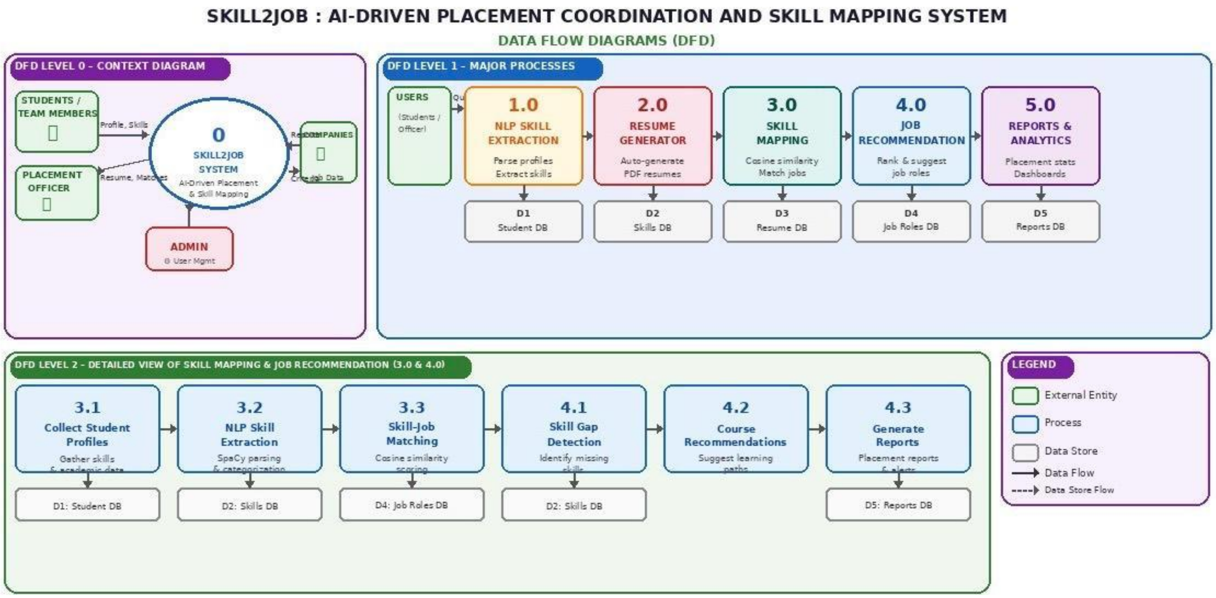

Figure 5.3: DFD Level 0, 1 & 2

This Figure 5.3 illustrates the Data Flow Diagrams (DFD) for the system across three levels, detailing how information moves between users, the AI-based skill mapping engine, and various databases for placement coordination and job recommendation.

DFD Level 0 – Context Diagram

The Level 0 diagram provides a high-level overview of the system. The Skill2Job System acts as the central entity interacting with four external entities:

- Students/Team Members provide profile details, academic information, technical skills, and queries to the system and receive generated resumes, job recommendations, skill-gap reports, and alerts.
- Placement Officer provides company data and eligibility criteria, and receives shortlisted candidate reports and placement analytics.
- Companies provide job role requirements and receive matched candidate profiles from the system.

- Admin manages user accounts, company records, system settings, and overall data management.

This level shows the overall interaction between external users and the system without exposing internal process details.

DFD Level 1 – Major Processes

Level 1 breaks down the system into major functional modules:

- NLP Skill Extraction Module (1.0) — Parses student profiles and uploaded resumes using SpaCy-based NLP to extract and categorize technical skills for downstream processing.
- Resume Generator (2.0) — Automatically generates professional PDF resumes from student profile data including skills, education, projects, and experience.
- Skill Mapping (3.0) — Computes compatibility scores between student skill vectors and company job requirement vectors using cosine similarity and keyword-based matching algorithms.
- Job Recommendation (4.0) — Ranks and delivers personalized job role and company suggestions to students based on computed compatibility scores; identifies skill gaps and recommends learning paths.
- Reports & Analytics (5.0) — Generates placement reports, skill distribution dashboards, and performance analytics for placement officers and administrators.

Each process interacts with corresponding data stores:

D1: Student Database

D2: Skills Database

D3: Resume Database

D4: Job Roles & Company Database

D5: Reports Database

DFD Level 2 – Detailed Job Recommendation & Skill Gap Process

Level 2 provides a detailed view of the Job Recommendation module (4.0):

4.1 Collect Student Profiles – Retrieves academic details, skills, and project data from the student database

4.2 NLP Skill Extraction – Parses and categorizes technical skills from profiles using SpaCy-based processing

4.3 Skill–Job Matching – Compares student skill vectors against job requirement vectors to compute compatibility scores

4.4 Skill Gap Detection – Identifies missing skills by comparing the student profile against target job role requirements

4.5 Course Recommendations – Suggests relevant online courses and learning paths to bridge identified skill gaps

4.6 Generate Reports – Produces ranked shortlists, placement analytics, and recommendation summaries for users and placement officers

The process interacts mainly with:

D2 (Skills Database)

D4 (Job Roles & Company Database)

D5 (Reports Database)

The DFD clearly shows how students interact with the system through skill submission and profile creation, how resumes are generated and skills are mapped to job roles, and how placement recommendations are continuously refined. The system ensures efficient data handling through multiple databases and provides timely job matches, skill-gap insights, and reports for better placement decision-making.

CHAPTER 6

WORK PLAN

Week No.	Timeline	Activity	Description
1	Feb 03 – Feb 09, 2026	Literature Survey & Research	Reviewed AI-driven placement systems, NLP-based resume tools, and skill mapping papers. Identified the problem statement and core objectives for Skill2Job.
2	Feb 10 – Feb 16, 2026	Requirement Analysis	Collected functional and non-functional requirements from students, placement officers, and admins. Defined key system features and constraints.
3	Feb 17 – Feb 23, 2026	System Design (High Level)	Designed system architecture, DFD (Level 0 & 1), use case diagram, and ER diagram. Finalized the database schema for all major modules.
4	Feb 24 – Mar 02, 2026	Technology Stack & Planning	Finalized tech stack: React, Python Flask, MySQL, and NLP libraries. Planned module breakdown, development approach, and work breakdown structure.
5	Mar 03 – Mar 09, 2026	UI/UX Design	Designed wireframes and prototypes for student, placement officer, and admin interfaces. Focused on responsive layout and intuitive navigation.
6	Mar 10 – Mar 16, 2026	Environment Setup	Configured Git repository, Flask backend, MySQL database, and IDE. Deployed the initial project structure to establish a stable development baseline.
7	Mar 17 – Mar 23, 2026	User Authentication Module	Developed user registration, login, password reset, and role-based access control for student, placement officer, and admin roles.
8	Mar 24 – Mar 30, 2026	Student Profile Module	Implemented student profile creation with academic details, skills input, project upload, and profile update functionalities.
9	Mar 31 – Apr 06, 2026	Admin & Placement Officer Module	Developed the admin module to manage student records, add company details, set eligibility criteria, and monitor placement progress.
10	Apr 07 – Apr 13, 2026	Resume Generator Module	Implemented automated professional resume generation from student profile data including skills, education, projects, and experience as PDF output.
11	Apr 14 – Apr 20, 2026	NLP Skill Extraction Module	Integrated SpaCy-based NLP to parse and categorize technical skills from student profiles and uploaded resumes for downstream skill analysis.
12	Apr 21 – Apr 27, 2026	Skill Mapping Algorithm	Implemented cosine similarity and keyword-based matching to compare student skill profiles against company job requirements and compute compatibility scores.
13	Apr 28 – May 04, 2026	Job Matching Engine	Developed an ML-based job matching engine that ranks and recommends suitable job roles for students based on computed skill compatibility scores.
14	May 05 – May 11, 2026	Skill Gap Analysis Module	Identified missing skills for target job roles and recommended relevant online courses and learning paths to improve student employability.
15	May 12 – May 18, 2026	Job Recommendation Module	Integrated the end-to-end recommendation pipeline to deliver personalized company and role suggestions based on student skill profiles.
16	May 19 – May 25, 2026	Analytics & Dashboard	Developed placement analytics dashboards with skill distribution charts, placement statistics, and candidate shortlisting reports for officers.
17	May 26 – Jun 01, 2026	Integration & API Testing	Integrated all modules and validated API endpoints for resume generation, skill mapping, job matching, and recommendation services.

Week No.	Timeline	Activity	Description
18	Jun 02 – Jun 08, 2026	System Testing	Performed unit, integration, and user acceptance testing using sample student data to validate system accuracy and performance.
19	Jun 09 – Jun 15, 2026	Bug Fixing & Optimization	Resolved identified bugs, optimized ML model performance, improved resume generation quality, and enhanced UI/UX responsiveness.
20	Jun 16 – Jun 22, 2026	Deployment & Documentation	Deployed the system on a cloud server and prepared the final project report, user manual, and complete technical documentation.

Table 6.1 Detailed phase-wise work plan

This Table 6.1 represents the timeline of the project activities across different phases. It ensures systematic development from requirement analysis to final documentation. It helps in tracking progress and timely completion.

The following Gantt chart outlines the planned timeline for the development and completion of the Skill2Job project across seven weeks.

Figure 6.1: Project Work Plan – Gantt Chart

■ : Batch Formation

■ : Problem Statement Selection & Guide Allotment

■ : Literature Survey & Review-I

■ : Requirement Analysis

■ : Review-II

■ : System Design

■ : Implementation

■ : Testing

■ : Documentation

■ : Final Review & Submission

Figure 6.1 represents the timeline ensures systematic progress from requirements analysis through to final report writing, with dedicated phases for design, development, and testing.

This Gantt chart outlines the timeline for the “AI-Driven Placement Coordination and Skill Mapping System” project from January to June 2026. It shows a structured progression starting with planning and research phases, followed by design, development, testing, and documentation. The chart ensures clear task scheduling and timely reviews, leading to final submission in June.

CHAPTER 7

EXPECTED OUTCOME

The expected outcomes of the Skill2Job system, as defined in the project objectives and validated through the PPT presentation, include:

- Automatic generation of professional resumes using student profile and skills — eliminating the need for manual resume writing and ensuring a consistent, industry-standard format.
- Efficient skill mapping with accurate job recommendations — the AI engine matches each student's skill set to the most compatible job roles, improving placement accuracy significantly compared to CGPA-based filtering.
- Identification of skill gaps to improve employability — the system highlights missing skills required for target job roles and recommends specific courses or learning resources to address those gaps.
- Faster shortlisting of candidates and improved placement process — placement officers gain access to a ranked, filtered list of suitable candidates for each company, drastically reducing the time spent on manual screening.

Beyond these direct outcomes, the system is expected to contribute to a measurable improvement in overall campus placement rates and student employability indices over successive academic years.

CHAPTER 8

CONCLUSION

The Skill2Job system provides an intelligent solution to modern placement challenges by integrating Artificial Intelligence and Machine Learning techniques. The platform simplifies placement coordination, improves candidate selection accuracy, and enhances student employability.

By automating resume generation, skill analysis, and job-role recommendation processes, the system reduces manual workload and improves placement outcomes for students, placement officers, and institutions alike. The implementation of Skill2Job contributes to the development of efficient recruitment systems and supports career growth for students.

The Skill2Job system aims to improve campus placements by integrating AI-based skill mapping and job recommendation techniques. By analyzing student skills and company requirements, the system can provide accurate candidate matching and skill-gap analysis, which helps both students and placement officers. This project will contribute to enhancing employability and optimizing placement processes in educational institutions.

Future enhancements may include integration with corporate recruitment APIs, LinkedIn profile parsing, online learning platform integration, and predictive analytics for placement success forecasting.

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